

CLAUDENET v3: Three DR10 Grade-A Lenses New to the Huang/Storfer/Inchausti Lineage — Rediscoveries of Known Strong Lenses

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Each is a previously-published lens (CASSOWARY, DES, AGEL) recovered independently by v3 at DESI resolution — external validation, not net-new discovery

1 Three grade-A lenses, new to the lineage — all rediscoveries

CLAUDENET v3’s DR10 contaminant-aware vetting (C-vet) surfaced three grade-A candidates that the *lineage* crossmatch — Storfer 2024, Inchausti 2025, and Huang 2020/2021 — reported as absent from the catalogued sample, i.e. “new” (Table 11 of `papers/main.pdf`). A literature-wide NED/SIMBAD crossmatch (`94_external_xmatch.py`), however, shows that each coincides to within $\leq 0.5''$ with a previously-published strong lens found by *another* survey. They are therefore independent **rediscoveries**, not net-new discoveries — a genuine external validation that v3 recovers real lenses at DESI resolution, but not new objects. (A fourth C-vet survivor, `s_310364_6649`, is the Huang+2020 grade-A lens `DESI-038.2078-03.3906`; it is handled in the main report.)

The cutouts below are the DECaLS grz imaging at the finder’s native geometry (101 px, $0.262''/\text{px}$), fetched and rendered with the group’s inspection-viewer renderer (Lupton $Q=8$, 400 px upsample) by `93_make_gallery.py` — the exact pixels the v3 model and the panel saw. For each object we show the *full* cutout, a $2.5\times$ center *zoom*, and a lens-light-subtracted *residual* that exposes the arc/ring.

Honest framing. The net-new count from this set is **zero**: all three (and the Huang+2020 object) are known lenses. The value is external validation, not discovery. These positions have *no* Euclid coverage, and net-new discovery at DESI’s $\sim 1.3''$ seeing remains resolution-bounded.

2 Cutout panels

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Table 1: The three v3 grade-A C-vet survivors that are new to the Huang/Storfer/Inchausti lineage, with the previously-published lens each rediscovers (literature NED/SIMBAD crossmatch, `94_external_xmatch.py`).

DESI id	RA (deg)	Dec (deg)	C-vet p_{lens}	rediscovered lens (survey)	sep
s_441355_1111	177.1381	+19.5009	0.74	CSWA 1 (CASSOWARY)	0.5''
s_124958_10481	58.1768	−38.4292	0.82	DES J0352−3825 (DES)	0.2''
s_102952_9916	334.8017	−43.8098	0.76	AGEL J221912−434835 (AGEL/DES)	0.2''

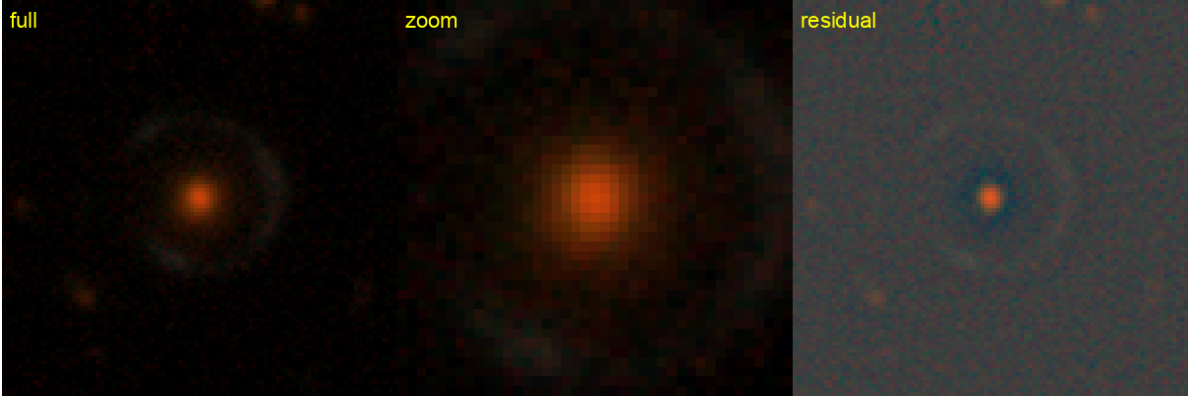


Figure 1: s_441355_1111 — RA 177.1381°, Dec +19.5009°; C-vet $p_{\text{lens}} = 0.74$. **Rediscovery of CSWA 1 — the Cosmic Horseshoe** (CASSOWARY survey; Belokurov et al. 2007), a near-complete Einstein ring around a massive LRG. Literature match 0.1–0.5'' (SIMBAD CSWA 1 and its lensed images). Panels: full / zoom (2.5×) / lens-light residual.

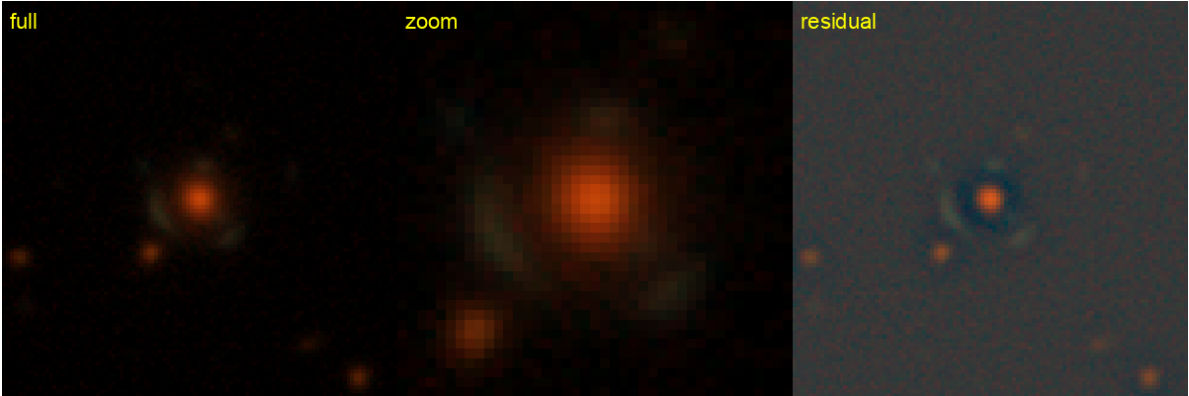


Figure 2: s_124958_10481 — RA 58.1768°, Dec −38.4292°; C-vet $p_{\text{lens}} = 0.82$. **Rediscovery of DES J0352−3825** (Dark Energy Survey; O’Donnell et al. 2022). Literature match 0.2'' to the lens galaxy; NED lists three lensed images at 2.6–3.7''. Panels: full / zoom (2.5×) / lens-light residual.

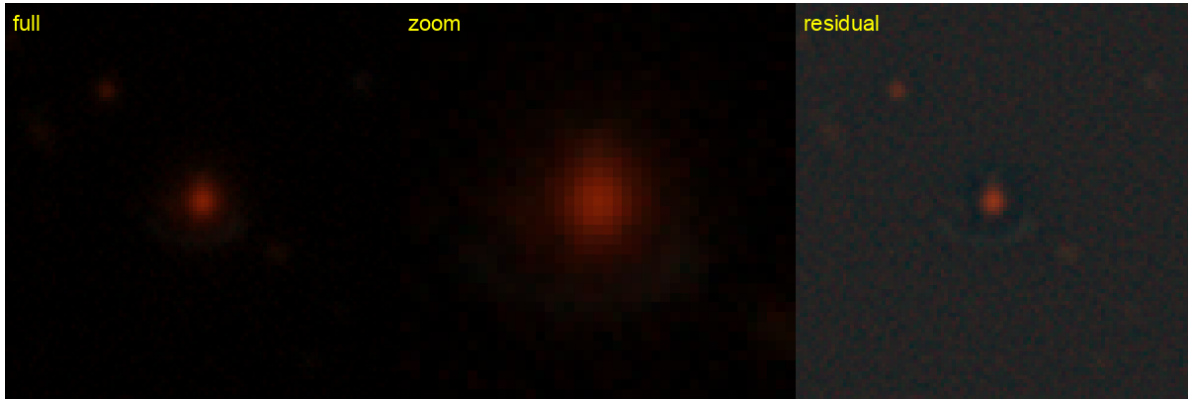


Figure 3: **s_102952_9916** — RA 334.8017° , Dec -43.8098° ; C-vet $p_{\text{lens}} = 0.76$. **Rediscovery of AGEL J221912–434835** (AGEL survey, Keck-confirmed; Tran et al. 2022; also DES J221912.39–434835.1). Literature match $0.2''$. The residual panel surfaces a near-complete blue ring around the red elliptical core. Panels: full / zoom ($2.5\times$) / lens-light residual.